

Galileo Galilei

Born in Pisa, Italy, in 1564, Galileo Galilei is best known for discovering evidence that planets orbit the sun, not Earth, but he was also a polymath—a scientist and inventor in many fields. Having first trained in medicine at the University of Pisa, Galileo is said to have become fascinated by physics after watching the pendulum-like motion of a swinging chandelier.

THROUGH THE TELESCOPE

After hearing reports of the newly invented telescope, Galileo built his own much improved versions around 1609. His telescope revealed the phases of the planet Venus and a family of moons orbiting Jupiter. These features convinced him that the heliocentric (sun-centered) model of the universe put forward in 1543 by Polish astronomer Nicolaus Copernicus must be correct.



Galileo's sketches of the moon



Large, heavy objects only fall more quickly than small, light ones because their ratio of weight to surface area is greater.

TESTING GRAVITY

Galileo studied the properties of falling objects and realized that, if it weren't for air resistance, bodies with different masses would fall toward the ground at the same rate. He is said to have demonstrated this to pupils around 1590 by dropping two things with different masses from the top of the Leaning Tower of Pisa.



► THE ITALIAN POLYMATH

This Italian portrait of Galileo shows him with the tools of his trade, including his telescope and a celestial globe on the table.

LIFE STORY

1564

Galileo is born in Pisa, to parents Vincenzo (a noted musician) and Giulia.

c. 1581

At the University of Pisa, he studies medicine but also takes up mathematics and natural philosophy, and applies mathematical analysis to the study of physics.

1592

Galileo becomes professor of mathematics at Padua. While there, he invents the thermoscope and a military compass.

1609

He builds his first telescopes and uses them to make observations of the skies.

